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Laser Synthesis of Bismuth Nanoparticles for Multimodal Theranostics of Oncological Diseases

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Abstract—The effect of laser radiation parameters on properties of bismuth (Bi) nanoparticles (NPs) produced by pulsed laser ablation in liquids (PLAL) was studied. We demonstrated that the average NPs size increases under the increase of laser radiation energy and the decrease in the distance from the focusing lens to the target surface. The obtained results provide a tool for a controllable synthesis of new functional nanomaterials for biomedical applications.

Keywords: laser ablation in liquids, size, nanoparticles, bismuth, theranostics

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1. INTRODUCTION

Methods of pulsed laser ablation in liquids (PLAL) have proved their efficiency for highly productive synthesis of ultrapure colloidal solutions of functional nanoparticles (NPs), which are widely used in biomedicine, catalysis, photovoltaics, and other applications [1, 2]. We recently applied PLAL methods for the synthesis of NPs, which demonstrate great potential as sensitizers of cancer therapies under external stimuli [3–7] and as efficient catalysts for biofuel cells [8]. PLAL is a physical synthesis route, whose advantages include simplicity, efficiency, high purity of obtained nanomaterials [9], and the possibility for synthesis of NPs with unique geometries and compositions of almost any inorganic material [10, 11]. Bismuth NPs produced by PLAL are considered as promising sensitizers for multimodal theranostics (therapy and diagnostics) of oncological diseases [12]. Being the heaviest non-radioactive chemical element, Bi has the highest X-ray absorbance [13], which makes Bi NPs efficient contrast agents in computed tomography (CT) [13] and sensitizers of X-ray capture therapy [14]. Furthermore, Bi NPs have high optical absorption in a wide spectral range, which makes them promising sensitizers of photohyperthermia and contrast agents in photoacoustic imaging [15].

Size characteristics of nanomaterials present the key parameter that determine NPs applicability in biomedicine as they affect the colloidal stability, toxicity, and the biodistribution of NPs [16]. However, the effect of laser parameters on size characteristics of Bi NPs produced by PLAL remains unexplored.

In this work, we study the effect of laser pulse energy and the distance from the focusing lens to the target surface on the average size and size dispersion of Bi NPs produced by the PLAL.

2. METHODS AND MATERIALS

Bi NPs were synthesized by femtosecond (fs) laser ablation in acetone, which was developed and described in detail in our previous studies [12]. The schematic representation of the experimental setup is shown in Fig. 1. A crystalline bismuth target (GoodFellow, USA, 99.999% purity) was fixed vertically in a glass cell (BK-7 glass) filled with 15 ml of analytical grade acetone. A 3-mm-diameter beam from a Yb:KGW laser TETA 10 (wavelength 1030 nm, pulse duration 270 fs, pulse energy up to 400 μJ, pulse repetition rate 100 kHz, Avesta, Russia) was focused by a flat-field (F-theta) objective with a 100 mm focal distance on the target surface through a side wall of the cell. The thickness of liquid layer between the input glass and the target surface was 3 mm. To increase the synthesis efficiency, the laser beam was moved over

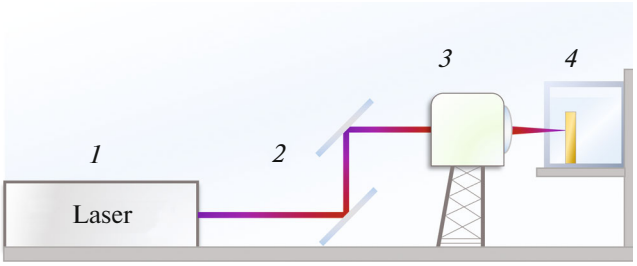

Fig. 1. Schematic representation of the experimental setup for PLAL. (1) fs Yb:KGW laser, (2) dielectric mirrors, (3) galvanometric scanner with a flat-plane objective, and (4) cell for ablation with target.

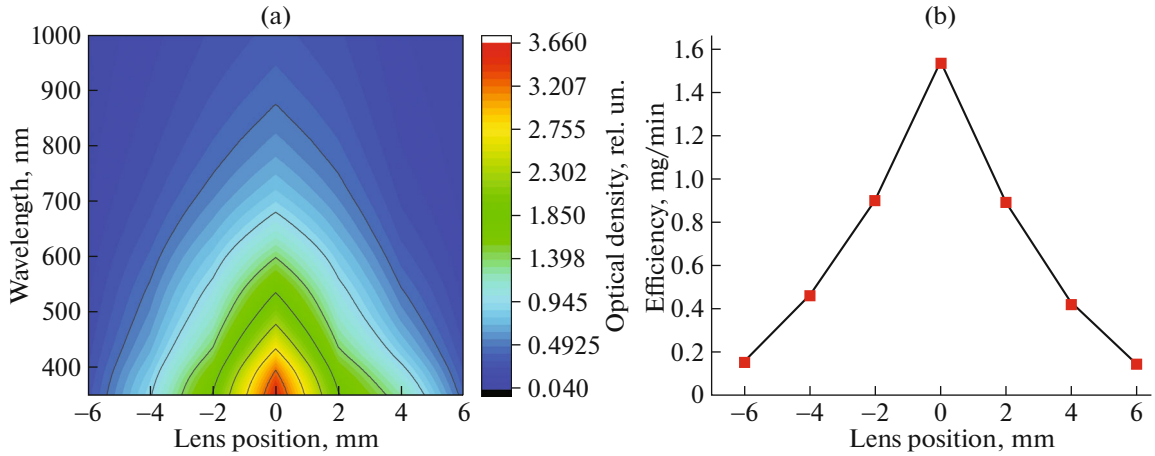

Fig. 2. (a) Dependence of the extinction spectra of Bi NPs colloidal solutions on the lens position relative to the target surface. (b) Dependence of the laser synthesis efficiency on the lens position relative to the target surface. The zero point is taken as the position of the maximum laser synthesis efficiency.

the target surface with the velocity of 5 m/s using a galvanometric scanner (LScan-10, Ateko-TM, Russia).

The morphology, size characteristics, and chemical composition of synthesized NPs were measured using a MAIA 3 (Tescan, Czechia) scanning electron microscope (SEM) aligned with an X-Act (OxfordInstruments, the Great Britain) energy-dispersive spectrometry (EDS) module. The hydrodynamic size was measured using a Zetasizer ZS (Malvern Instruments, France) by the dynamic light scattering (DLS) method. Optical extinction spectra of the obtained colloidal solutions of Bi NPs were measured in the spectral range of 330–1000 nm using a spectrophotometer ML 122 (SOL Instruments, Belarus) in glass cells with an optical path of 10 mm.

3. RESULTS AND DISCUSSION

The assessment of focal length during the propagation of high-intensity radiation in condensed media is complicated by the involvement of nonlinear optical effects. Therefore, the focal position in fs PLAL is commonly determined as the lens target distance at which maximum NPs productivity (mg/min) is observed. In this study, the focal length was determined by optical and gravimetric methods. In the optical method, extinction spectra of Bi NPs colloidal solutions obtained at various lens-to-target distances were measured (Fig. 2a). The spectrum with the maximum extinction at 330 nm corresponded to the proper focusing conditions. The data obtained by the optical method were independently confirmed by mass measurements of the dried NPs (Fig. 2b).

The change in the lens-to-target distance also affected the size characteristics of produced NPs. The average (mode) size of the distribution and its width decreased as this distance was increased (Fig. 3a). Such a dependence of size properties of synthesized NPs can be explained by the fact that

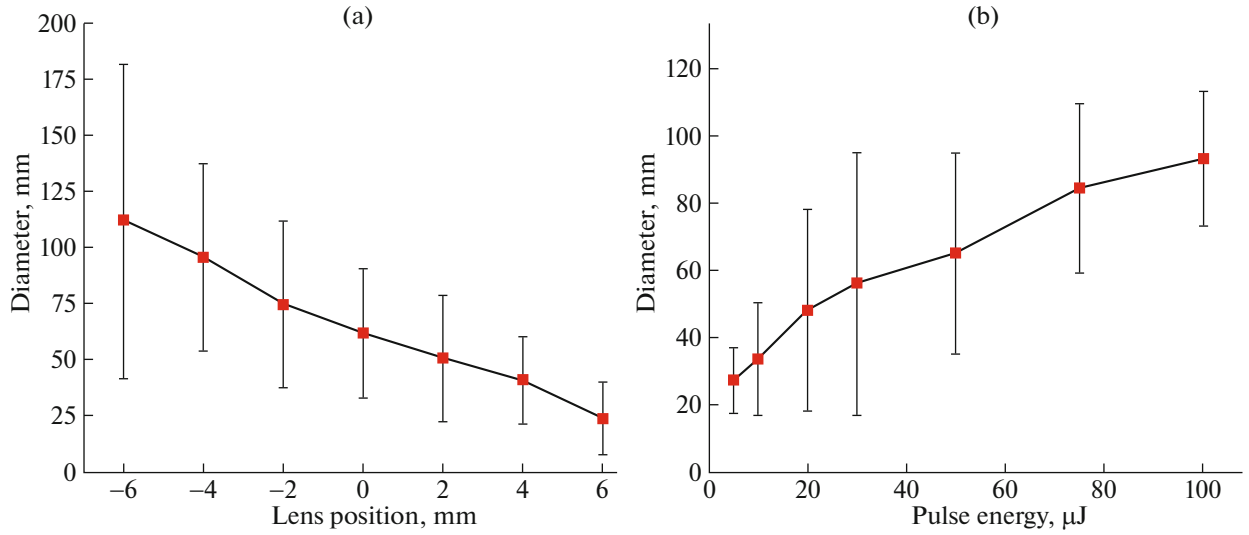

Fig. 3. Dependence of the mode (red squares) and size distribution width of Bi NPs (vertical lines) on the lens position relative to the target surface under the pulse energy is 20 μJ (a) and the laser pulse energy (b). The data were obtained by the DLS method.

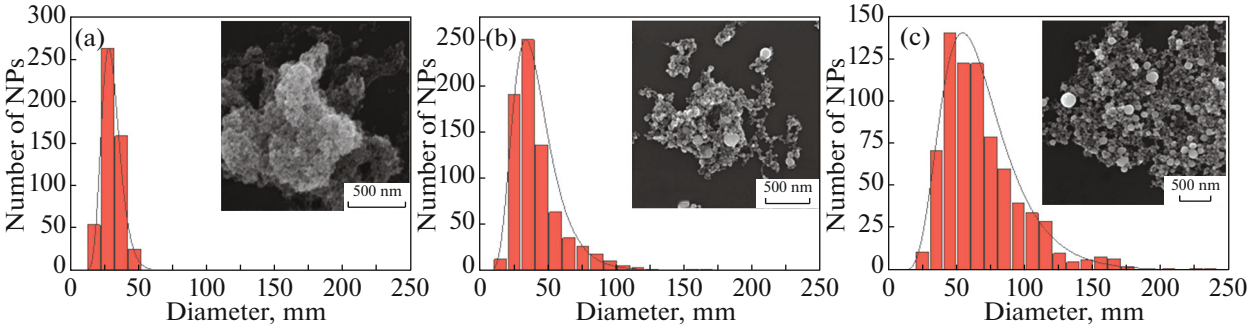

Fig. 4. Typical SEM images and size distributions of Bi NPs, obtained by the PLAL method with lens positions relative to the target surface (a) +6 mm, (b) 0 mm, and (c) –6 mm (the pulse energy is 30 μJ).

focused laser radiation causes an optical breakdown in the liquid in front of the target when lens-to-target distance is increased beyond the focal distance. Thus, a part of laser energy is lost, being absorbed, and scattered by the plasma formed in a liquid in front of the target during the optical breakdown. As a result, a smaller energy fraction reaches the target surface, which, as shown below, results in a decrease in the average size of produced NPs.

The dependence of the NPs size on the laser pulse energy was determined using a DLS method. These measurements revealed a decrease in the average size while the pulse energy decreases (Fig. 3b). The variation of average energy from 5 to 100 μJ led to the increase of the average NPs size from 25 to 95 nm.

The dependence of the average size and width of the size distribution on the lens-to-target distance was also studied by SEM imaging of the produced NPs (Figs. 4a–4b). The obtained data correlated well with the DLS data.

An optimal NPs size for the projected *in vivo* biomedical applications is about 50 nm [17]. This size provides efficient cell internalization and allows the NPs to avoid rapid removal from the bloodstream. Our experiments showed that that PLAL can provide this size of Bi NPs at laser energies of 30 μJ corresponding to maximal productivity of the method.

It should be noted that biomedical applications require the use of aqueous colloidal solutions of NPs. However, pure Bi NPs exhibit fast polymorphic transformation in an aqueous medium and change their shape from spherical to "flat flakes" [12]. As a result, they lose their colloidal stability and therefore they cannot be used for in vivo applications. This problem can be solved by coating the NPs with a biocompat-

ible polymer, which protects them against the polymorphic transformation at neutral pH, as it was shown previously [12]. However, the polymorphic transformation of the NPs at low pH remains possible, which can be used to deliver NPs to cell cytoplasm via the endocytic pathway.

Thus, laser-synthesized Bi NPs is very promising nanomaterial for biomedicine with many various applications.

4. CONCLUSIONS

Colloidal solutions of Bi NPs were synthesized by the PLAL in acetone. The influence of the laser pulse energy and the lens-to-target distance on the size of the obtained nanomaterials was studied. It was shown that the selection of laser synthesis parameters (average energy, lens-to-target distance) makes possible the production of Bi NPs having optimal sizes for biomedical applications. The obtained data can be used to obtain Bi NPs with desired properties for applications as multimodal theranostic sensitizers.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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